Stones, Obstruction & Trauma

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 93	Management of Upper Urinary Tract Obstruction
Ch 94	Upper Urinary Tract Trauma
Ch 95	Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis
Ch 96	Acute Care of Urinary Lithiasis
Ch 97	Surgical Management of Ureteral Calculi
Ch 98	Surgical Management of Upper Urinary Tract Calculi
Ch 99	Prevention of Urinary Stone Disease
Ch 100	Lower Urinary Tract Calculi
Ch 54	Management of Paediatric Kidney Stone Disease
Ch 125	Genital and Lower Urinary Tract Trauma
Ch 62	Paediatric Genitourinary Trauma
★	Putting it together

93 Management of Upper Urinary Tract Obstruction

1 · Ureteropelvic junction obstruction (UPJO): presentation

- Trauma → *Ch 94, Ch 125*; stones → *Ch 95–100*.
- Usually congenital. **Neonates**: prenatal hydronephrosis on ultrasonography; azotaemia if bilateral or solitary. **Older children/adults**: intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI), rarely hypertension.

2 · Diagnosis

KEY POINT

Dilation is not obstruction. Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows *where*; functional tests (renography, Whitaker) show *whether it matters*. Combine one of each.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula (vs multicystic kidney); ureter not seen
Non-contrast CT (acute pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study
Retrograde pyelography	**At the time of repair** (avoids infection)
Nephrostomy + antegrade study	Retrograde fails; tests recoverability; permits Whitaker

- P1 Radionuclide studies and diuretic renography in PUJ obstruction — *tracers, well-tempered protocol, T½ cut-offs, P40, O'Reilly curves*
- **Renography (Ch 93 values)**: hydrated → furosemide **0.3–0.5 mg/kg** at **20 min**; MAG3 preferred. **Obstructed: T½ > 40 min**; after successful pyeloplasty drainage occurs even before furosemide.

ALGORITHM

Whitaker test: nephrostomy → bladder catheter → perfuse at **10 mL/min** → non-obstructed ≤12–15 cm H_2O; obstructed > **15–20 cm H_2O**. Invasive → only when other tests non-diagnostic. Errors: pelvic compliance, position

3 · Indications and choice of intervention

- **Indications**: symptoms; impaired or falling function; stones or infection; rarely hypertension.
- **Neonates**: selective observation — operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults**, indeterminate → serial renography.
- **Procedure of choice**: dismembered pyeloplasty, now robotic at most major centres; endopyelotomy fails more.
- Pre-repair drainage only for infection, azotaemia (solitary/bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**

Endopyelotomy subgroup (landmark series)	Success
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- Crossing vessel is **prognostic, not a contraindication**, but pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO — cross over**: failed endopyelotomy → pyeloplasty; failed pyeloplasty → endourology first.
- **Nephrectomy**: thin shell + normal contralateral kidney. Adults **<15%** differential function = non-salvageable; children — no agreed cut-off. Uncertain → temporary stent/nephrostomy, re-scan. Also stones with chronic infection, multiple failed repairs.

4 · Endopyelotomy

- **Principle**: full-thickness **lateral** incision into peripelvic fat, healing over a stent; lateral is **free of crossing vessels**.
- Pyeloplasty (≈**95%**) wins: wide dependent funnel and pelvic reduction; endourology wins only on recovery.
- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy; marked vessel entanglement on 3D CT → robotic. Sterile urine first; if obstruction prevents clearance → temporise with stent/nephrostomy.

Percutaneous antegrade

- Best for **UPJO + pelvicalyceal stones**. Puncture **mid-posterior or superolateral calyx**, not lower pole (acute infundibular angle); upper pole risks hydropneumothorax. Working + safety wire across UPJ before incising.
- Stent: 14/7-Fr endopyelotomy stent (or 8–10 Fr); traditionally **6 weeks** (2 vs 4 weeks equal).
- **Clear stones before incising** (fragments migrate periureterally → fibrosis). Pelvic stone may cause oedematous pseudo-UPJO → clear, then reassess.

Retrograde ureteroscopic (standard)

- P1 Retrograde endopyelotomy — *indications, holmium lateral-incision steps, results (≈90% without crossing vessel >4 mm), complications, balloon/cautery-wire alternatives*

> **ALGORITHM**
> General anaesthesia → wire + safety wire → flexible ureteroscope via sheath → 200-µm holmium fibre → incise **laterally** into fat, into normal ureter → stent + Foley → renogram 4 weeks after stent removal

- **Percutaneous endopyeloplasty**: endoscopic Heineke–Mikulicz; fails in secondary UPJO.

Complications and follow-up

> **ALGORITHM**
> Percutaneous bleeding: bed rest, transfusion → **do not irrigate** nephrostomy (tamponade) → persistent → **selective embolisation** (low threshold)

- Renogram 1 month after stent removal → 6 months → yearly for **2–3 years**. Pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- Failure → CT angiography; significant crossing vessel → **pyeloplasty, not repeat incision**.

5 · Pyeloplasty

KEY POINT

Principles (open = minimally invasive): widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** transition.

- P3 P1 Laparoscopic pyeloplasty; Management of failed pyeloplasty — *Anderson–Hynes steps, approaches, contraindications, stent 4–6 weeks, vascular hitch, late failures, redo options*
- MCQ Trendelenburg 1886 first attempt; Kuster 1891 first successful dismembered; **Anderson–Hynes 1949** standard; Davis 1943 intubated ureterotomy.
- Dismembered: any insertion, pelvic reduction, **transposition over crossing vessels**; only technique that excises the abnormal UPJ. Poor for long proximal strictures or small intrarenal pelvis.

ALGORITHM

Dissect ureter with periureteral tissue → lateral marking stitch → excise UPJ → spatulate **laterally** → apex to inferior pelvis → transpose anterior to vessel → watertight absorbable anastomosis (± reduction) → stent → drain

- **Stent** out at **4–6 weeks**. Added nephrostomy in uncomplicated adults → longer stay, more infection; reserve for secondary UPJO.
- Laparoscopic/robotic success >**90–95%** (= open); robot's advantage is ergonomic. "Strict" success definitions ≈one-third lower.

ALGORITHM

Foley out 24–36 h → drain out if negligible; output rises after Foley removal → **replace Foley 7 days** (reflux along stent) → stent out 4–6 weeks → renography

6 · Other UPJ reconstructions

KEY POINT

Every UPJ repair asks: **what tissue bridges the gap while keeping blood supply?** Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width ≤**3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- **Y-V**: V base on dependent medial pelvis, apex at UPJ; Y stem down lateral ureter. Flaps: outline longer than needed; trim apex, never base.
- **Intubated ureterotomy**: soft stent + nephrostomy; at 6 weeks nephrostogram → stent out → clamp → nephrostomy out.

- P2 Uretero-calycostomy — *resect lower-pole parenchyma (not nephrotomy), mucosa-to-mucosa, omental cover; cortex <5 mm predicts failure*

7 · Retrocaval ureter

- P3 Retrocaval ureter — *venous anomaly, always right, fish-hook/S-shape, CT; transpose ureter anterior to IVC*

8 · Ureteral stricture — evaluation and endourology

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis, endoscopy, infection (TB, schistosomiasis).
- **Exclude malignancy first** (goblet sign; metastases) → CT ± ureteroscopy with biopsy.
- Stricture after ureteroscopy (URS) ≈ **3%** vs 1.5% after SWL; impacted stone >2 months → ≈1 in 4. Thulium > holmium stricture rate.
- Endourology needs ipsilateral function ≥ **25%** (algorithm 20%).
- **Stents**: intrinsic strictures; extrinsic compression → tandem or metallic stent (changed yearly). Chronic stenting for unfit/cancer patients.
- **Balloon dilation**: 4-cm, 5–8-mm balloon until **waist disappears**, 10 min → stent 2–4 weeks. Contraindicated: infection, stricture > **2 cm**. Favourable: short, recent (≤3 months), non-anastomotic, well vascularised (≈85%); anastomotic ≈50%; transplant strictures respond well.
- **Endoureterotomy** (retrograde holmium, first line <2 cm): full thickness, **2–3 mm normal ureter** each side; safety wire; larger stents (8–12 Fr) better.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels
Proximal ureter	**Lateral/posterolateral**	Great vessels

- **Complete obliteration** (high-risk only): antegrade + retrograde scopes, **cut toward the light** → stent 8–10 weeks.

ALGORITHM
<2 cm + function >20% → endoscopic incision | >2 cm + >20% → reconstruction | <20% → nephrectomy (children may still merit repair). Transplant → consider balloon. Renography for 2 years

KEY POINT
Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · Ureteral stricture — reconstruction ladder

Define nature, site, length; exclude carcinoma; be ready for the next rung. Iatrogenic injury → *Ch 94*.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim

Technique	Defect bridged	Level
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy**: short defects only; preserve adventitia; spatulate 5–6 mm 180° apart; double-J; omental cover. Drain out after 24–48 h minimal output (check drain creatinine). Success >90%.

ALGORITHM
Persistent leak: X-ray stent position → drain off anastomosis → stop suction → Foley + anticholinergic → nephrostomy

- P3 P2 Boari flap; Trans Uretero Ureteric anastomoses — *contracted bladder contraindicates; recurrent stricture commonest; eGFR fell in 44%; TUU donor above IMA, contraindications, correct recipient reflux*
- Refluxing = non-refluxing reimplant in adults; robotic = open success, shorter stay. Treat outlet obstruction first.
- **Renal descensus**: kidney in Gerota fascia rotated inferomedially, lower pole fixed; limited by renal vein.
- **Ileal ureter**: isoperistaltic; contraindicated by creatinine >**2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Distal ileum ≥15 cm from ileocaecal valve; segments >15 cm prevent reflux. Tapering unproven. Dilation + acidosis → seek outlet obstruction; endoscopic surveillance from year 3.
- **Buccal mucosa onlay** (robotic, omental wrap): dense scar. **Autotransplantation**: contralateral kidney absent/poor or other repairs impossible.

KEY POINT
Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · Ureteroenteric anastomotic stricture

- Conduits 4–9%; ≈5% after 10 years → **lifelong upper-tract surveillance**.
- **Cause = ischaemia** (adventitial supply). **Left commoner**: extra length, under sigmoid mesentery, angulation at IMA.
- Risk: **non-refluxing anastomosis** (≈13% vs 2–3%), carcinoma in situ at margin, obesity, ASA >2, Clavien–Dindo 3–4.
- Conduit dilation often non-obstructive → renography. Irregular mass + rapid progression → recurrent cancer.
- **Open revision = gold standard** (≈91% vs 26% endoscopic); robotic equal. Endourology **antegrade** → balloon or holmium → internal–external stent (flushable). Success <1 cm ≈50% vs >1 cm 6%; right ≈41% vs left 19%.

ALGORITHM
<2 cm + >20% → right: endoscopic incision; **left: formal repair** (sigmoid mesentery bleeding) | >2 cm → formal repair | <20% → nephrectomy or observation

11 · Retroperitoneal fibrosis

- P3 P1 Retroperitoneal fibrosis — *idiopathic 70% vs secondary (drugs, lymphoma); plaque at L4–L5, medial deviation; PET for activity; deep biopsies; drain → prednisone (80% respond) → bilateral*

ureterolysis + omental wrap

ALGORITHM

Uraemia → **nephrostomy** or stents → manage **post-obstructive diuresis** → stop culprit drug, biopsy → steroids → ureterolysis

- Steroids act while lesion is cellular: pain and ESR fall within days. Second line: immunosuppressants, **tamoxifen**. Ureterolysis starts at distal non-dilated ureter, "split and roll"; omental wrap most definitive.

94 Upper Urinary Tract Trauma

- P2 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — *mechanism, haematuria, AAST 2018, imaging indications, CT red flags, non-operative care, exploration, vascular control, renovascular injury, complications*

1 · Renal injury — mechanism and presentation

- Commonest urological organ injured; blunt 80–90% (motor vehicle collision commonest); most grade I–III, heal conservatively.
- **Deceleration** injures fixation points (hilum, UPJ): artery thrombosis, vein disruption, pedicle/UPJ avulsion.
- **Penetrating**: more bleeding, nephrectomy, complications. High-velocity missiles → blast, cavitation → delayed necrosis. Stab at anterior axillary line → hilum/pedicle; posterior axillary line → parenchyma. Mark gunshot entry/exit with metallic markers.
- Signs: flank ecchymosis, **ipsilateral rib 9–12 fractures** (strongest association).

2 · Haematuria

- Significant: **gross**, or **microhaematuria (>5 RBC/hpf) with SBP <90 mm Hg**. Degree ≠ severity; absent in ~⅓ blunt renovascular and **~½ of UPJ injuries**.
- Test the **first aliquot**. Dipstick false positives: myoglobinuria, menses, povidone-iodine.

> **KEY POINT**
> Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · AAST kidney injury scale (2018)

Grade	AAST 2018
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration **≤1 cm**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's**
IV	Collecting system laceration with extravasation; pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's; segmental/complete infarction without active bleeding
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for bilateral/multiple injuries up to III; grade by highest finding. MCQ
  Trap: main-vessel injury was IV in 1989, now V; segmental pseudoaneurysm/AVF now IV.

4 · Indications for imaging (AUA = EAU)

- Penetrating — **any haematuria**, if stable for CT.
- Blunt with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 at any time.
- Blunt microhaematuria without shock/deceleration → **observe without imaging**.

- **Children**: larger kidney, less fat, pliable ribs; catecholamines hold BP until ~50% blood loss → image liberally. UPJO lets minor trauma cause major injury.

5 · Imaging studies

- **Contrast CT with arterial/nephrographic and delayed (~10 min) phases** = gold standard. Renal vein injury poorly seen (medial haematoma with normal arteries suggests it).
- Omit delays if kidneys normal and no perinephric/pelvic fluid; if suspected and none obtained → repeat urographic phase.
- FAST: poor specificity; cannot separate haematoma from urine.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Contrast blush	Brisk bleeding → strongly predicts AE
≥2 of: rim distance >4 cm, medial laceration, blush	High-risk grade IV

- **MiGUTS nomogram** (bleeding intervention): penetrating, blush, pararenal haematoma, associated injuries, shock; largest contributor **haematoma rim distance**.
- **One-shot IVP** (unstable, no CT): **2 mL/kg contrast → single film at 10 min** — confirms a functioning contralateral kidney. Alternatives: palpate it; loop injured ureter + IV methylene blue.

Perinephric collection	HU	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · Nonoperative management

- Stable, staged **grade I–IV and selected V** → expectant, any mechanism; >90% success in blunt even with extravasation. **Early mobilisation is safe**.
- **Failure triggers**: instability; transfusion of **2–3 units**; new pain/guarding; enlarging urinoma with fever. Most failures need only stent, nephrostomy or AE.
- **Penetrating** breaches Gerota's (no tamponade) → more failures; stab > gunshot. Stable stab posterior to posterior axillary line → serial examination.
- Failure within 24 h rare (~3%): higher grade, penetrating, nonrenal abdominal injury.
- **Reimaging** (AUA): CT ~48 h for **grade IV–V deep lacerations** or deterioration; not uncomplicated I–III.

7 · Urinary extravasation

- Parenchymal leaks: **76–90% heal**. **Drain early** when urine cannot pass antegrade: pelvis/UPJ avulsion, ureteric clot, UPJO, parapelvic cyst.
- Symptomatic/enlarging urinoma → retrograde stent + catheter → add nephrostomy or drain. Retrograde impossible (urethral injury, transection, no lithotomy) → nephrostomy ± antegrade stent.

- >25% nonviable parenchyma with extravasation → more abscess; lower threshold to operate.

> **KEY POINT**
> Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · Intervention

Absolute indications	Relative (increasingly observed)
Instability, no/transient response	Extravasation with significant devascularisation
Expanding or pulsatile haematoma	Colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

- **AE**: superselective, raises salvage; repeat AE succeeds; needs experienced team, no delay. **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria.
- At laparotomy, exploration **not mandatory** for staged I–III; only absolute indication **pulsatile/expanding haematoma** (opening Gerota's releases tamponade).

> **ALGORITHM**
> Eviscerate small bowel → incise peritoneum over aorta above IMA → to ligament of Treitz → aorta to left renal vein → loop vein, retract → loop both arteries → reflect colon → open Gerota's → occlude if bleeding

- **Damage control** (hypothermia, acidosis, coagulopathy): pack → temporary closure → ICU → return 24–48 h; may permit AE instead of nephrectomy.

> **ALGORITHM**
> Exposure → vascular control → limited debridement → suture-ligate vessels → **watertight collecting-system closure** → parenchyma over bolster → Gerota's or **pedicled omentum** → drain

- Colon/pancreatic injury no longer mandates nephrectomy.
- **Revascularise** only solitary or bilateral injury, warm ischaemia **<120 min**; unilateral salvage only ~⅓. **Left renal vein may be ligated; right cannot**. Hypertension after thrombosis repair ≥43% → monitor BP a year. Endovascular stent of intimal flap: anticoagulation is the drawback.

9 · Complications and follow-up

- AKI relates to age, transfusion, polytrauma rather than nephrectomy. Delayed bleeding usually within 21 days → AE.
- **Page kidney** (subcapsular compression): ACE inhibitor/ARB; refractory → drainage or decortication. Goldblatt = artery stenosis; post-traumatic AVF → AE.
- EAU: no imaging for grade I–II; salvaged III–V → DMSA. Return to sport 2 weeks–3 months; bleeding risk peaks **2–6 weeks**. Solitary kidney does not bar contact sport.

10 · Ureteral injury

- P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — *aetiology, risk sites, prevention, CT with delays/retrograde, damage control, repair by level (hitch 95–100%, Boari up to 12 cm, TUU, ileal ureter), follow-up*

AAST ureter grade	Description
I	Contusion/haematoma without devascularisation
II	Laceration **<50%** transection
III	Laceration ≥50% transection
IV	Complete transection, **<2 cm** devascularisation
V	Avulsion, >2 cm devascularisation

- Iatrogenic commoner overall (distal); external mostly gunshot (proximal). **Haematuria absent 25–45%**.
- **Distal ureter** commonest iatrogenic site; minimally invasive hysterectomy riskier than open; prolapse surgery highest gynaecological risk; endometriosis raises risk. ~10% of women: distal supply mainly from uterine artery → devascularisation presenting at 1 week–2 months (stenosis, urinoma, ureterovaginal fistula).
- **Ureteroscopy**: stricture ~3%, avulsion <1%. Stop basketing at a tear and stent; holmium safest; ≤14-Fr sheath in unstented ureter; safety wire. Partial nephrectomy/ablation of lower-pole/hilar tumours → late proximal stricture (pyeloperfusion protects). **Ureteroarterial fistula** (chronic stent + pelvic surgery/radiation) → endovascular covered stent.
- Preoperative stents aid **recognition**, not proven prevention. Intraureteral indocyanine green fluoresces for hours. Cystoscopy with efflux check after minimally invasive hysterectomy and prolapse surgery.
- Efflux aids: oral phenazopyridine (most cost-effective), IV indigo carmine (safe), fluorescein, dextrose irrigation. **Methylene blue** = MAO inhibitor (serotonin toxicity), avoid in pregnancy/G6PD deficiency; falsely lowers oximetry.

Modality	Role
CT urography, **10-min delays**	Best in stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic — never rely on it

- CT signs: **no contrast distal to injury**, medial extravasation (UPJ avulsion), delayed nephrogram, urinoma. ~11% missed at laparotomy.

KEY POINT
Ureteral injury is diagnosed by absent contrast beyond it — so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

ALGORITHM
Stable at laparotomy → repair now · unstable → damage control, delayed repair · contusion → stent or excise and repair · late or partial → stent → fails → nephrostomy, delayed repair · ureterovaginal fistula → stent first

- **Damage control**: ligate proximal stump with long silk → nephrostomy once stable; or exteriorised single-J/5-Fr feeding tube. Open nephrostomy too slow.
- Principles: spare adventitia; debride to bleeding edges; **spatulated, tension-free, stented, watertight**; drain. **No bowel or autotransplantation acutely.** Longitudinal laceration → close transversely.

- Trauma reimplant **non-tunnelled** (tunnelled strictures more). Psoas hitch preferred over distal ureteroureterostomy (tenuous stump supply); suture to psoas minor tendon, spare genitofemoral/femoral nerves.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

1 · Overview and epidemiology

- **Removal is surgical, recurrence is metabolic**. Prophylaxis *→ Ch 99*.
- Lifetime prevalence **1–15%**; US ≈ **10%** and rising (partly CT detection).
- **Sex**: male 2–3 : 1, gap narrowing (steepest rise in adolescent females). **Oestrogen protective** (↑ renal calcium reabsorption, ↓ bone resorption) → second female peak at menopause.
- **Race**: highest White, lowest Asian. **Age**: peak 4th–6th decades; paediatric stones rising 5–10%/year.
- **Climate**: hot, arid; US Southeast "stone belt"; summer peak. Heat → **low urine volume**; sunlight → **vitamin D hypercalciuria**.
- **Occupation**: heat → low volume and pH → uric acid; sedentary/weightlessness → hypercalciuria; restricted bathroom access → low volume.
- **Obesity/BMI**: independent risk, stronger in females. **Metabolic syndrome**: risk rises with number of traits. **Diabetes** bidirectional.
- **Insulin resistance** unifies: low urine pH → uric acid stones; hyperinsulinaemia → hypercalciuria.
- Hypertension, MI, stroke, CKD, ESRD associated (causation unproven). Fluid intake inversely related (proven by RCT); water hardness weak effect.

2 · Physicochemistry and stone formation

- P2 Stone formation (etiopathogenesis) and medical management of urolithiasis — *saturation states, heterogeneous nucleation, steps, Randall plaque vs ductal plugs, inhibitor list, matrix 2.5%*
- Cascade: **supersaturation → nucleation → growth/aggregation → retention → stone**. CP = product of ion concentrations; Ksp = CP at saturation; Kf = CP of spontaneous crystallisation.

State	CP	What happens
Undersaturated	< Ksp	No crystals; stones may dissolve
Metastable	Ksp–Kf (most urinary salts)	No homogeneous nucleation; growth and heterogeneous nucleation possible; **inhibitors effective — zone of therapy**
Unstable	> Kf	Spontaneous nucleation; inhibitors ineffective

- **RSR** = CP ÷ Ksp (EQUIL 2 overestimates; JESS matches direct measurement): < 1 dissolve, > 1 grow.
- pH change lowers saturation of **phosphate, uric acid, cystine — not oxalate** (oxalic acid pK ≈ 4). **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- Urine holds calcium oxalate at ≈ 4× aqueous solubility; with inhibitors precipitates only at **7–11×**. Nephron transit 5–7 min: aggregation, not single crystals, occludes tubules. Free- vs fixed-particle (Randall plaque anchor) theories.
- **Randall plaque** (MCQ 1937): **calcium apatite** from **thin-limb basement membrane** → interstitium → erodes papilla → calcium oxalate grows attached; osteopontin at mineral–

organic interface. Vascular hypothesis: vasa recta calcification.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present — anchor	Uncommon, mild
Intestinal bypass	Absent	Apatite plugs in inner medullary collecting ducts; injury, fibrosis
Brushite	Present	Apatite plugs; interstitial fibrosis
Distal RTA, hyperparathyroidism, cystinuria	—	Present

KEY POINT

Plaque vs plug: plaque = idiopathic interstitial apatite anchor; plug = intratubular obstruction with injury from secondary systemic disease. **Ductal plugs → search for a secondary cause.** Infection or SWL may convert a calcium oxalate former into a brushite former.

3 · Inhibitors and promoters

- Inhibitors raise the supersaturation needed for nucleation. **No inhibitor of uric acid or cystine** — pH and concentration only. Promoters: low volume, low citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits precipitation, agglomeration, urate-induced nucleation	Most important small-molecule inhibitor
Magnesium	Complexes oxalate	Synergy with citrate, phytate
Glycosaminoglycans	Bind crystal surface	Lower in stone formers
Nephrocalcin	Inhibits COM nucleation, growth, aggregation	Formers excrete isoforms lacking γ-carboxyglutamic acid
Tamm-Horsfall protein	Inhibits aggregation, not growth	Most abundant urinary protein (TAL); **promoter in acid urine**
Osteopontin	Nucleation, growth, aggregation, epithelial binding	Inducible

- **Matrix** ≈ **2.5%** of stone weight; matrix substance A (formers only).

4 · Mineral metabolism

- **Calcium**: 30–40% absorbed (small bowel); transcellular (calcitriol-dependent, saturable) at low intake, paracellular at high. Luminal oxalate/phosphate/fatty acids bind calcium — **oxalate-rich food ↓ calcium absorption and luminal calcium ↓ oxalate absorption**.
- **Calcitriol** (proximal tubule 1α-hydroxylation, stimulated by **PTH and hypophosphataemia**) = most potent gut calcium stimulator. **PTH**: bone resorption; ↑ renal calcium reabsorption, ↓ phosphate reabsorption, ↑ calcitriol; **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive; claudin-16/19; CaSR → claudin-14 blocks
DCT	8–10% (fine control)	Transcellular: TRPV5 → calbindin-D28k

- Only 1–3% of filtered calcium excreted.
- **Oxalate**: 6–14% absorbed, unregulated, excreted almost entirely in urine; half dietary, half hepatic (glyoxylate precursor). **Ascorbic acid** largest endogenous source (up to 40%); hydroxyproline ≈ 15%. *Oxalobacter formigenes* degrades oxalate and stimulates colonic secretion.

5 · Classification

Stone	Frequency
Calcium oxalate (pure or with apatite)	**60%**
Hydroxyapatite	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite	≈ 7% (falling)
Cystine	1–3% (up to 10% children)
Triamterene, silica, 2,8-DHA	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. phosphate leak)	↑ Gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	PHPT	3–5
Hyperuricosuric calcium	Purine excess	10–40
Hypocitraturic calcium	Idiopathic, diarrhoea, distal RTA, thiazide	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	↓ Gut magnesium	5–10
Idiopathic low urine pH	Low pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease; cystine transport	1–5; < 1
No abnormality	—	< 3

KEY POINT

Classify by the **urinary abnormality, not the stone**: composition starts the work-up. Fewer than 3% have no abnormality. Work-up → *Ch 99*.

6 · Hypercalciuria

- Commonest abnormality (35–65%); ↑ calcium salt saturation, ↓ inhibitor activity.

- **Definition**: > **200 mg/day** after 1 week of 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day; or > 7 (men) / 6 (women) mmol/day.
- **Pak classification** (gut, kidney, bone) — exam framework, not used to direct therapy.

Feature	Absorptive	Renal	Resorptive
Defect	↑ Gut absorption	Renal leak	PHPT
Serum calcium	Normal	Normal	**High**
PTH	**Suppressed**/normal	**High** (secondary)	High
Fasting urinary calcium	**Normal** (< 0.11 mg/dL GF)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption → PTH suppressed → ↑ filtered load	Renal loss → PTH ↑ → calcitriol ↑	PTH → bone resorption + calcitriol → hypercalcaemia

- Absorptive **type I** persists on calcium restriction; **type II** normalises.
- **CYP24A1** loss (cannot inactivate calcitriol) → infantile hypercalcaemia; adults hypercalciuria, nephrocalcinosis — worse with vitamin D.
- **Renal phosphate leak** → hypophosphataemia → ↑ calcitriol → ↑ gut calcium (SLC34A1/A3).

Disorder	Gene	Features
Dent disease (X-linked)	ClC-5	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, renal failure
FHHNC	Claudin-16, -19	TAL magnesium/calcium loss
Bartter	NKCC2, ROMK, ClC-Kb, barttin; type V activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria

KEY POINT

Fasting urine + PTH separate gut from kidney: renal leak → fasting calcium high, PTH high; absorptive → fasting normal, PTH suppressed.

- **PHPT** ≈ 5% of stones; stones in < 5% of PHPT. Suspect if serum calcium > **10.1 mg/dL**; repeat with intact PTH; ionised calcium for normocalcaemic PHPT. **Thiazide challenge** unmasks PHPT (hypercalcaemia without PTH suppression). PTH also causes phosphaturia and mild hyperchloraemic acidosis. Others: thyrotoxicosis, vitamin D toxicity.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High/inappropriately normal	Commonest in outpatients
Sarcoidosis	Macrophage 1α-hydroxylase	Suppressed	Corrects with **corticosteroids**
Malignancy	PTHrP, osteolysis (RANKL)	Suppressed	Commonest in inpatients
Glucocorticoid excess	↑ Bone resorption	Raised	Cushing syndrome

7 · Hyperoxaluria

- **Definition**: > **40 mg/day**. Primary, enteric, dietary, idiopathic.

- P2 Primary Hyperoxaluria — *PH1–3 enzymes, markers, course; oxalate ≥100 mg/day, oxalosis; pyridoxine (Gly170Arg); lumasiran; liver–kidney transplant in PH1*
- Exclude PH in relatives before living-related donation.

ALGORITHM
Fat malabsorption → fatty acids saponify calcium/magnesium → more free oxalate → **colonic hyperabsorption** (bile salts ↑ permeability)

- **Enteric** (commonest acquired): small-bowel resection/Crohn, jejunoileal bypass, cystic fibrosis, malabsorptive bariatric (Roux-en-Y, duodenal switch — not band/sleeve). **Needs intact colon**; ileostomy → normal oxalate but uric acid/ammonium urate stones. Diarrhoea also → hypokalaemia, hypocitraturia, hypomagnesiuria, low pH.
- **Dietary**: nuts, chocolate, tea, spinach, rhubarb. **Severe calcium restriction counter-productive**; vitamin C ↑ oxalate. Less *O. formigenes* in formers; antibiotics deplete it. Mild idiopathic hyperoxaluria may matter as much as hypercalciuria (Arabian Peninsula).

8 · Hyperuricosuria, hypocitraturia, hypomagnesiuria

- **Hyperuricosuria** > **600 mg/day** (diet, gout, myeloproliferative, HGPRT). Urate promotes calcium oxalate by **inhibitor adsorption, salting out, epitaxy**; allopurinol cut recurrence (trial below).
- **Hypocitraturia** < **320 mg/day**; **acid–base status is main determinant**. Causes: distal RTA, chronic diarrhoea, animal protein, thiazides (hypokalaemia), ACE inhibitors.

KEY POINT
Citrate excretion reads proximal-tubular **intracellular pH**: acidosis, hypokalaemia or acid diet make the tubule reclaim citrate — hence potassium citrate → *Ch 99*.

- Hypomagnesiuria: **two RCTs of magnesium — no benefit**. **Low urine pH** (< 5.5) = former "gouty diathesis" → uric acid and calcium oxalate stones.

9 · Renal tubular acidosis

- P2 Renal tubular acidosis — *type 1/2/4 comparison (urine pH, K^+, citrate, stones up to 70%), incomplete dRTA, ammonium chloride load, topiramate/acetazolamide*

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**	< 5.5 at steady state	Low
Urinary citrate	**Profoundly low**	Near normal	—
Stones	**Up to 70%** — calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Associations	Sjögren, lupus, obstruction; genetic	Fanconi	Diabetic nephropathy

- **Incomplete distal RTA**: normal electrolytes but urine pH cannot fall < 5.5 after ammonium chloride load.
- Distal RTA: stones lead to diagnosis in up to half; **profound hypocitraturia** probably the most important factor. Genetics: SLC4A1 (AE1, dominant); ATP6V1B1/ATP6V0A4 (recessive,

sensorineural hearing loss). Type 2 self-limiting; citrate spared.

> **KEY POINT**
> Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria — the recipe for **calcium phosphate** stones.

10 · Uric acid stones

- P2 Uric acid calculi: etiopathogenesis, medical management and allopurinol — *no uricase, pKa 5.35, three determinants (low pH dominant), insulin resistance/ammoniagenesis, idiopathic vs hyperuricosuric calcium*

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, diarrhoea, gout	**Single most important**
Low volume	Diarrhoea, heat	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative, uricosurics	Least often the driver

- Humans lack hepatic **uricase** (allantoin 10–100× more soluble). Sodium urate ≈ 20× more soluble than free acid.
- Uric acid pKa **5.35**; undissociated solubility ≈ 96–100 mg/L; at pH 6.5 > 1200 mg/L soluble. Radiolucent on KUB, visible on NCCT. BMI ≥ 27 predicts urine pH ≤ 5.5.

> **ALGORITHM**
> Insulin resistance / renal lipotoxicity (fatty acids not glutamine; ↓ NHE3) → ↓ ammoniagenesis → H^+ as titratable acid → **persistently low urine pH** → undissociated uric acid → stone

> **KEY POINT**
> Uric acid stones are a disease of **urine pH, not uric acid**. No inhibitor exists, so raising pH above the pKa removes the driver; †alkalinisation can dissolve stones → *Ch 99*.

11 · Cystine and struvite stones

- P1 P4 Cystinuria; Struvite stones — *SLC3A1/SLC7A9 types, solubility by pH, hexagonal crystals, CKD; urease cascade, biofilm, Proteus/Ureaplasma, discordant cultures*
- Cystinuria: rBAT/SLC3A1 (2p, type A), $b^{0,+}$AT/SLC7A9 (19q, type B), both (type AB); ornithine, lysine, arginine also lost but soluble. Incidence 1 in 1000–17,000; first stone ≈ 12 years; males more severe; coexisting hypocitraturia commonest; ~¼ adults eGFR < 60. †Hexagonal crystals; cyanide–nitroprusside screen.
- Cystinuria: autosomal recessive, only cystine stones form; **no inhibitor**; solubility meaningful only at alkaline pH (≈ 1000 mg/L at pH 9).
- Struvite: ***Proteus mirabilis*** commonest; *S. aureus*, **Ureaplasma** (easily missed); *E. coli* rarely urease-positive → bladder and stone cultures discordant. 5–15% of stones, female 2 : 1; elderly, diabetes, stasis, diversion, spinal cord injury.
- Struvite: urease absent from sterile urine; bacteria encased in stone → complete clearance. **Less than half of staghorns are infection stones** (most calcium phosphate). †"Coffin-lid" crystals.

> **ALGORITHM**
> Urea → (urease) $NH_3 + CO_2 \rightarrow NH_4^+ + OH^-$ → pH 7.2–8.0 → with magnesium: **struvite**; with calcium: carbonate apatite

12 · Miscellaneous and drug-related stones

	Xanthine	2,8-DHA
Enzyme deficient	XDH	APRT
Radiology	Both radiolucent — mistaken for uric acid	
Treatment	—	Allopurinol or febuxostat (more effective)

- **Ammonium acid urate**: radiolucent; diarrhoea, IBD, ileostomy, **laxative abuse**, UTI; endemic paediatric bladder stones.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	Insoluble at physiological pH; stones 4–13%; †radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicates	Drug stones
Indirect	Acetazolamide, **topiramate**, zonisamide	Distal RTA-like → calcium phosphate
Indirect	Loop diuretics	Hypercalciuria (up to 64% of LBW infants on furosemide)
Indirect	Steroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, urate

13 · Anatomic predisposition and pregnancy

Anomaly	Stones	Metabolic findings
UPJO	≈ 20%; children ~70× risk	Common (hypercalciuria first); recurrence after repair, often contralateral
Horseshoe kidney	≈ 36% adults	All tested abnormal; **hypocitraturia**; calcium ≈ 89%
Calyceal diverticulum	Up to 40%	Abnormal in half; UTI up to 40%
Medullary sponge kidney	Frequent + nephrocalcinosis	Hypercalciuria, hypocitraturia; dRTA on acid loading

> **KEY POINT**
> Anatomy sets the stage but metabolism writes the stone: anatomical stone formers still need **metabolic evaluation**; pyeloplasty alone does not prevent recurrence → *Ch 99*.

- **Pregnancy**: 1 in 204–3000, 2nd–3rd trimester. Physiological hydronephrosis up to 90%, **right > left**, resolves 4–6 weeks post partum. GFR ↑ and placental calcitriol promote; citrate/magnesium inhibit — **net risk unchanged**, stones skew to **calcium phosphate**.

14 · Landmark trials

Trial	Design	Result	Take-home
Borghi water trial	RCT, high fluid vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier → *Ch 99*
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones → *Ch 99*

96 Acute Care of Urinary Lithiasis

1 · Presentation and diagnosis

- **Renal colic**: severe flank pain to groin, patient writhes. UVJ stone: tip-of-penis pain, frequency. Haematuria ~**90%** (absent >10%). **Fever → infected obstructed system** → urgent intervention.
- Bloods, urinalysis, **pregnancy test** in women of reproductive age. Urine >50 WBC/hpf or nitrites suggest UTI (raised blood WBC may be stress).

Modality	Role	Limitation
NCCT	Gold standard (sens 97%, spec 95%); alternative diagnoses	8–16 mSv; misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm, obesity
Ultrasound	First line in pregnancy, children, known formers	Up to 40% still need CT
KUB	**Avoid** for acute diagnosis	Lowest accuracy
MRI	Second line in pregnancy/children	Poor stone detection

- STONE score failed validation as NCCT substitute.

> **KEY POINT**
> Unknown diagnosis → NCCT answers everything at once. Known stone → US showing hydronephrosis suffices.

2 · Pain management

- Obstruction → stretch → prostaglandins → diuresis, ↑ intrarenal pressure, spasm. **NSAIDs** treat the cause; equal to opioids with less vomiting and rescue.

Agent	Place	Note
Ketorolac 30 mg	**First line** (EAU, NICE)	Avoid in renal failure, GI bleed, pregnancy
IV paracetamol	Second line	NSAID barred/insufficient
Opioids	Intractable pain	**Continue NSAID** — morphine + ketorolac beats either
Antispasmodics; forced fluids	**Ineffective**	Fluids only for vomiting

- Opioid dependency rises with each day beyond day 3.

3 · Medical expulsive therapy (MET)

- P2 Medical expulsive therapy (MET) in urinary calculi — *α1 mechanism, tamsulosin/silodosin, passage-by-width table, SUSPEND/STONE, benefit in distal 5–10 mm, intervene after 4–6 weeks*
- Trial ± MET **4–6 weeks**. Measure **transverse width**: 6 mm ≈ 33% pass; mid-ureter ≈ 60%.

4 · Urgent intervention

- **AUA**: obstruction + infection; bilateral obstruction; solitary kidney; immunosuppression; renal deterioration; refractory pain or vomiting. Uninfected obstruction harms only if prolonged (>~2 weeks).
- **Stent or PCN** equal for sepsis; stent → dysuria/haematuria; PCN → longer stay.
- **Stent failure**: P-CUT (ureteric diameter at stone − stone width) ≥ **7.5 mm** (strongest); stone ≥9.5 mm; multiple → prefer PCN.
- Acute SWL (proximal, <24 h) or URS (distal <10 mm) **only without infection** → *Ch 97*.

ALGORITHM

Colic → analgesia + NCCT → infection/AUA criterion? Yes → drainage only (stent/PCN) → definitive later. No → stone <10 mm → trial ± MET 4–6 weeks → fails → URS/SWL

5 · Special populations

- P2 Stones (urolithiasis) in pregnancy — *US → MRI → low-dose CT, conservative first, stent/PCN encrustation, URS any trimester*
- Pregnancy: hydronephrosis below iliac vessels suggests stone; resistive index difference ≥ **0.06** = obstruction. ~30% need intervention; change stent/PCN every **4–6 weeks**; early gestation favours PCN; preterm delivery highest with PCN. **URS safe any trimester** (ACOG 2019).

Population	Pitfall	Key point
Elderly	Atypical pain; first stone >60 unusual	Always CT — exclude aortic/cardiac mimics; early drainage
Transplant kidney	Denervated → **painless**	Fever/renal failure; endoscopic, else percutaneous

6 · Landmark trials

Trial	Design	Result	Take-home
SUSPEND	RCT, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm

97 Surgical Management of Ureteral Calculi

- P2 Management of upper ureteric stones — *passage rates, EAU choice by size, SWL vs URS table, SWL failure factors, flexible URS, antegrade and laparoscopic routes, complication rates*

1 · Background and indications

- Dangers: **renal damage and sepsis**. NCCT gold standard (low-dose equal); US misses mid-ureter; US + KUB for follow-up once localised (radiolucent stones invisible).
- MET controversial after SUSPEND *→ Ch 96*; intervene after **4–6 weeks**. Ladder: SWL → URS → PCNL with antegrade URS.

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis/sepsis → **drain only**, never manipulate the stone

- Work-up: blood count, creatinine, **urine culture**; exclude coagulopathy, untreated UTI. Preoperative KUB confirms radio-opacity.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high risk
SWL	Stop if feasible	Not recommended — choose another modality
Laser URS	**Need not stop**	None; more haematuria, stent symptoms

KEY POINT
Why only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen and uncontrollable; laser URS works intraluminally under vision.

2 · Choosing treatment

- Upper ureter ≤2 cm: similar 3-month SFR, more SWL retreatment. Distal: similar SFR, SWL more sessions, fewer complications. **Impaction** (thick wall on NCCT) defeats SWL → endoscopy. Large hard proximal → PCNL + antegrade URS (unfit → staged URS).

Modality	Size guidance
SWL	Historical limit 2 cm; SFR falls above **10–15 mm**
URS	No definite limit
PCNL ± antegrade URS	>15–20 mm, large hard proximal

KEY POINT
Constant trade-off: SWL = fewer complications, more sessions; URS = one session, slightly more morbidity.

3 · Ureteroscopes

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered	Smallest tip	~8.5 Fr	7.7 Fr
Cost	Cheap	Repair ~every **15** cases		Low volume

- Semirigid: distal/mid ureter; damaged by torque at brim. Flexible: proximal stones (follows migration). Digital **not more durable**; keep a fibreoptic scope for narrow ureter. Raised IRP → fever, sepsis.

4 · Access

- P3 Accessories for RIRS; Ureteric stents and ureteroscopy with or without stenting — *guidewire types, 12/14 Fr sheath, safety wire (Segura), pre-stent 2–4 weeks, stent indications, Denstedt stentless trial*
- Consent: access failure, further procedures, perforation, avulsion, sepsis, stricture, nephrectomy.
- **Safety wire** advised (complexity unpredictable). Difficult ureter → stent, return after **2–4 weeks** (≥14 days for SFR; ≤4 weeks for lower sepsis).

> **ALGORITHM**
> Wire will not pass: open-ended catheter 1–2 mm short of stone → RPG → contrast hydrodissection + hydrophilic wire → advance catheter past stone → exchange for stiff wire

- Catheter abutting stone → wire exits at 90° → **perforation**. ~1 in 10 ureters inaccessible → **stent and stage**.
- Scope will not enter orifice: **railroading** between two wires, dual-lumen catheter, rotate flexible scope, obturator.

Dilation method	Best for
Indwelling stent	Whole ureter narrow — gentlest; commits to second procedure
Serial/coaxial dilators	Long narrow segments
Balloon	Focal stenosis — until waist resolves, 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting not needed** for ureteral stones (benefit only renal). **α-Blockers ~1 week pre-URS**: easier access, higher SFR, less stent pain.
- **Sheath**: for burden, dusting, high power; lowers IRP and heat. Reserve for stones **above the iliac vessels**. 14/16 Fr ↓ ureteral blood flow (stricture rare).

5 · Lithotripsy

- **Basketing**: stone **5–10 mm from tip**; **never blind**; when in doubt **fragment first**. Entrapped: push to wider segment → disassemble basket → scope alongside → laser stone → else cut basket.

Source	Key points
Electrohydraulic	High ureteral trauma; abandoned
Pneumatic	~12 Hz; **rigid/semirigid only**; best below brim; retropulsion
Ho:YAG	Short pulse fragments; long/modulated pulse ↓ retropulsion; dusting = low energy, high frequency
Thulium fibre laser	Low peak power, long pulse → fine dust; minimal retropulsion; >40 Hz misses

- P4 Thalium laser in urology — *TFL physics, settings (0.6 J × 6 Hz; dust 0.2 J × 40 Hz), "paint, do not drill", RCTs vs holmium*

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	0.6 J × 6 Hz SP (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- Antiretropulsion devices improve SFR with semirigid URS. Impacted stone: target centre first, sides fall away from wall. **Settings, not only the laser, decide outcome**.
- **Thermal**: damage above **43 °C**; **heat tracks power, not laser type**. Cap **≤10 W in ureter**, ≤25 W kidney; sheath, continuous irrigation (stopping it → thermal stricture).

6 · After URS

- **Single perioperative IV antibiotic dose**; extend only after urosepsis this episode.
- Stent symptoms ~4/5; stentless uncomplicated URS safe (yet >half stent). String removal preferred. Stent for perforation, stricture, tight ureter, infection, second stage; after sheath → inspect ureter on withdrawal.

7 · SWL for ureteral calculi

- Non-invasive, no GA or stent; option at all levels. Acute obstructing stone: SWL **within 48 h** beats delayed.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (treatment + 48 h); distal obstruction; untreated UTI; uncontrolled hypertension	Skeletal deformity/obesity preventing focusing; adjacent aneurysm

- **Predictors**: size most important (URS better ≥10 mm); impaction possibly strongest ("ring" sign, static stone); cystine, brushite, COM resist; HU >1000 and SSD >10 cm proven for renal only.
- **No routine pre-stenting**. Hold aspirin/NSAIDs/anticoagulants treatment + 48 h. Antibiotics only with stents/tubes, struvite, recent UTI.
- Technique → *Ch 98*: upper ureter supine, mid-ureter over sacrum **prone**, distal supine; **60–90 shocks/min** beats 120; ramp energy. Little gain beyond 2 sessions; salvage URS after failed SWL not harder. α-Blockers ↑ SFR; **forced diuresis useless**.

8 · Special situations

- **Pregnancy**: conservative first; **SWL contraindicated**; URS safe with shielded fluoroscopy; stents encrust → change **every 4 weeks**; retrograde stent before nephrostomy.
- **Diversion/reimplant**: SWL (prone) or **antegrade flexible URS**. Transplant: both routes safe. Anomalies: escalate SWL → PCNL → laparoscopic ureterolithotomy.

9 · Complications

- **Perforation** (0–4%): stop → stent ~4 weeks → drain urinoma. Extruded stone: leave it, inform patient. Submucosal stone: laser free + stent. Further lithotripsy embeds fragments → granuloma → stricture.
- **Avulsion** (0.06–0.5%): forceful extraction, **proximal third → nephrostomy + delayed repair**, avoid nephrectomy.

- **Stricture** 3–6% → image every patient (silent). **Sepsis** ~5%: positive culture, pre-stent, diabetes, long surgery → treat culture, keep IRP low.

> **KEY POINT**
> Nearly every serious URS complication is force without vision — blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

- Future: robotic URS; **burst wave lithotripsy** (handheld, clinic-based).

10 · Landmark trials

Trial	Design	Result	Take-home
Stentless URS	RCT, stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

98 Surgical Management of Upper Urinary Tract Calculi

1 · Choosing the approach

- Framework: 2016 American Urological Association (AUA)/Endourological Society guideline — index patients, shared decision making. Pathogenesis → *Ch 95*; ureteral stones → *Ch 97*.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction/stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe/ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure/transplant, solitary kidney, diversion, pregnancy

KEY POINT

Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude *a* modality, not surgery.

Asymptomatic renal stones

- Truly asymptomatic = no symptoms, no sequelae (incl. no recurrent urinary tract infection [UTI]). **~50%** have a stone event within **5 years**.
- **Size predicts intervention, not symptoms** (risk rises >5 mm and >10 mm).
- **Recurrent UTI + non-obstructing stone**: remove regardless of composition; **culture the stone**.
- Residual fragments **>4 mm** after ureteroscopy (URS) → treat.
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side (see trials).
- Declined treatment → medical prevention → *Ch 99* + serial imaging.

Preoperative evaluation

- AUA: non-contrast computed tomography (NCCT) **should** precede percutaneous nephrolithotomy (PCNL); **may** guide shock wave lithotripsy (SWL) vs URS.
- SWL best: density **<900 HU**, skin-to-stone distance (SSD) **<10 cm**.
- Urinalysis/culture for all (within 1 month).

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL fewer complications; URS higher stone-free rate (SFR)
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line — fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—

Scenario (AUA)	Offer	Caveat
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- European Association of Urology (EAU): >20 mm → PCNL, then URS/SWL; **10–20 mm → SWL or endourology in equipoise**; <10 mm → SWL or URS, then PCNL (lower pole 10–20 mm separate).

Stone morphometry scores

- P1 Stone complexity scores — *Guy's, STONE, CROES, S-ReSC predict SFR, not complications; Triple D predicts SWL*

2 · Shock wave lithotripsy

- P2 ESWL: mechanism, indications, factors affecting outcome, complications and current status — *peak **30–100 MPa**; sources; mechanisms (dynamic squeezing strongest); **60–90 shocks/min**, ramping, coupling; haematoma, steinstrasse (no routine stent); refuted effects; burst wave*
- First human SWL 1980 (Dornier); electrohydraulic HM3 = efficacy benchmark MCQ. Electromagnetic: consistent up to 10^6 shocks. Electrohydraulic focus drifts up to 1 cm.
- **Absolute contraindications**: pregnancy; uncorrected coagulopathy/antithrombotics; untreated UTI; severe untreated hypertension; aortic/renal artery aneurysm (aorta >**4 cm**); obstruction distal to stone; cannot target.
- No prophylactic antibiotic if urinalysis negative and no risk factors (AUA, EAU).
- **Arrhythmia**: gate shocks to cardiac refractory period. †Obstructed/infected steinstrasse → nephrostomy or URS.

3 · Ureteroscopy (retrograde intrarenal surgery, RIRS)

- Landmarks MCQ: Bozzini's Lichtleiter 1806 (first endoscope); Takayasu 1974 first ureteral access sheath (UAS).
- Urine test before elective URS; single-dose antibiotic per antibiogram; pre-treat bacteriuria.
- **Antithrombotics**: URS is the modality for those who must continue; anticoagulation appears safe, antiplatelets increase admissions.

Instruments and principles

- **Safety wire**: AUA/EAU recommend (EAU: controversial); multicentre study no outcome difference. Never force a wire past an impacted stone.
- **UAS**: lowers intrarenal pressure, eases repeated passes — **no clear fall in infection**. Ureteral injury in nearly half (severe ~1 in 8), less if **pre-stented**; de novo stricture rare.
- **Anti-retropulsion** devices (Stone Cone, NTrap, BackStop) reduce migration.
- **Stent omission** (AUA) when all four: normal anatomy, normal renal function, no ureteral injury, no planned second stage.

Success

"Clinically insignificant fragments" concept is flawed — small fragments cause reintervention in up to a fifth; report zero-fragment rates.

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes → treat urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (underestimated)
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter → immediate nephrostomy, delayed reconstruction

4 · Percutaneous nephrolithotomy

- Landmarks MCQ: Fernström and Johansson 1976 first percutaneous stone removal; Wickham 1984 first tubeless; Valdivia-Uría 1987 first supine; Jackman 1998 first mini-perc (13 Fr).
- Higher SFR, fewer procedures than SWL/URS for 1–2 cm; treatment of choice for **staghorn** and most >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture/injury/reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative

- Stop anticoagulants with prescriber (bridge if high risk); **low-dose aspirin is safe**.
- **NCCT is the gold standard**: anatomy, infundibular length/angle, SSD, HU, and ribs, pleura, spleen, liver, colon.
- Poorly functioning kidney: renogram — cut-off debated at 10–20%; nephrectomy may suit.
- Single-dose culture-directed prophylaxis. **Tranexamic acid** (Cochrane): less transfusion, no thromboembolic rise — individualise.

Positioning

- P1 P2 Prone vs Supine PCNL / patient positions in PCNL — *prone: upper pole access, ocular/brachial plexus risk; supine: airway, gravity drainage, ECIRS*

Access — the most important step

- Urologist-obtained access → shorter stay, fewer transfusions and secondary procedures.
- **Ideal**: posterior calyx, short wide infundibulum. Upper pole posterior reaches most of the system but **supracostal (above 12th rib)** raises chest complications. Lower pole: more colonic injury (left); steep infundibulopelvic angle hampers UPJ.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle end-on	Mediolateral in oblique view, cephalocaudal in anteroposterior (AP)
Depth	Rotate back past AP	Short advances alternating views

- **Air** fills non-dependent calyces but blocks ultrasound. Confirm by urine return, not contrast.
- **Ultrasound access**: no radiation; shows bowel, spleen, liver, pleura; stones by shadow and twinkle artefact.
- **Endoscopic combined intrarenal surgery (ECIRS)**: ureteroscope watches puncture → fewer tracts, less bleeding, higher clearance.

Dilatation and tract size

- Alken metal (dense scar), Amplatz serial, or **balloon** (one pass, 20–30 atm). Safety wire unless through-and-through.
- Multiple tracts raise bleeding, reduce function (esp. chronic kidney disease).
- P4 Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques — *standard 24–34, mini 15–22, ultra-mini 11–14, micro <10 Fr; mini: equal SFR, less bleeding*

Completion and exit

- **Flexible nephroscopy/URS** at the end — randomised trial: higher SFR, fewer tracts, less bleeding.

> **ALGORITHM**
> Confirm system and ureter clear → withdraw sheath under vision, safety wire last → compress tract → persistent bleeding: re-inflate balloon or **Kaye tamponade catheter** → alert anaesthetist → **selective angioembolisation**

- Drainage: nephrostomy (≤10 Fr = less pain), double-J stent, or tubeless; fibrin glue does not reduce bleeding.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications**, fewer secondary procedures.
- Haemorrhage (†delayed: arteriovenous fistula/pseudoaneurysm → angioembolisation), visceral injury, hydro/pneumothorax (†chest drain if significant), sepsis, gas embolism, death.

5 · Special anatomy

Anatomy	Key adaptation
Horseshoe (stones ~1 in 5)	SWL: long SSD. PCNL: access medial and caudal, upper pole posterior calyx usually subcostal; more colonic injury; frequent UPJO
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper/interpolar (laser-incise or dilate ostium); lower pole: PCNL. Fulguration controversial
Pelvic kidney	SWL/URS <2 cm; PCNL with ultrasound ± CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable
Transplant	Always remove stones; denervated graft → fever, oliguria, haematuria, not colic. Ultrasound PCNL
Spinal deformity	SWL poor; URS good; PCNL equal clearance, more infection

- **Laparoscopic/robotic**: only when SWL, URS and PCNL fail or are expected to; with UPJO, clear stones then reassess.

6 · Landmark trials

Trial	Design	Result	Take-home
Lower Pole I	randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

99 Prevention of Urinary Stone Disease

1 · Stone disease as a chronic condition

- Urinary stone disease is a **chronic metabolic disorder punctuated by colic**. Pathogenesis *→ Ch 95*.
- Recurrence up to **50%** within 5–10 years. **ROKS** nomogram predicts a second stone (ROKS 2: multiple). Key risk: **low urine volume** (single most important); also young, male, obesity, family history, low citrate.

2 · Screening and metabolic evaluation

- P1 P2 Metabolic Evaluation in Stone Disease — *screening (history, blood, urine, stone analysis, imaging) for every first-time former; high risk → one or two 24-hour urines; fast/calcium-load testing discouraged*
- 24-hour urine is **not** part of screening. Lithogenic drugs: calcium, vitamins C and D, acetazolamide, steroids.

3 · General measures (all formers)

- **Fluid** — cornerstone. Urine ≥**2.5 L/day** (≈3 L intake).
- **Limit nondairy animal protein**: raises urinary calcium and uric acid, lowers citrate.
- **Low sodium**: sodium raises calcium excretion, lowers citrate; Dietary Approaches to Stop Hypertension (DASH) diet lowers risk.
- **Normal calcium** — restriction *increases* stones. More fruit and vegetables.

> **KEY POINT**
> Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · Hypercalciuria — thiazides

- **Preferred drug** (American Urological Association, AUA): hypercalciuria with recurrent calcium stones, **and** recurrent calcium stones with normal evaluation. Yendt 1966 MCQ.
- **Hypokalaemia → intracellular acidosis → hypocitraturia**: add **potassium citrate**; check potassium 1–2 weeks after starting/changing dose.
- Amiloride + thiazide: more calciuric effect — **never add potassium**. Avoid triamterene (stones).
- Higher bone density; side effects: gout, diabetes, sexual dysfunction.

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	Gastrointestinal (GI) upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea; ataxia, numbness
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

5 · Hyperoxaluria

- Risk rises continuously with oxalate. Avoid large portions of spinach, nuts, chocolate, tea.
- **Enteric** (malabsorption, diarrhoea, Roux-en-Y gastric bypass): urine oxalate >40, often 60–70+ mg/day; fat saponifies calcium, freeing oxalate for colonic absorption.
- Enteric treatment: **calcium up to 1 g/day with meals**, very high fluid, **liquid** potassium citrate (wax tablets pass unabsorbed), magnesium, low oxalate.
- P2 Primary Hyperoxaluria — *oxalate often >**100 mg/day**; PH1 (~70%, AGXT, pyridoxine-responsive), PH2 (GRHPR), PH3 (HOGA1); lumasiran silences glycolate oxidase (ILLUMINATE-A)*
- PH3 mildest, often hypercalciuria. Lumasiran approved for PH1.

6 · Hypocitraturia

- **Potassium citrate** 30–60 mEq/day: AUA offers for recurrent calcium stones with low citrate **and** with no abnormality. Sodium citrate does not lower urinary calcium.
- Caution in renal impairment, potassium-sparing diuretics.
- **pH ≥7 → calcium phosphate** risk. **Alkali citrate, not total citrate, matters**; orange juice also raises oxalate; Moonstone supplement acidic MCQ.

7 · Uric acid stones — a pH disease

- P2 Uric acid calculi: etiopathogenesis, medical management and allopurinol — *pKa **5.5**; low pH the main cause; potassium citrate to **6.0–6.5**, avoid >7.0; allopurinol not first-line; hypersensitivity (fatal up to 27%)*
- Sodium alkali risks calcium stones. Allopurinol only for hyperuricaemia with hyperuricosuria or failed alkalinisation; risk higher with thiazides.

8 · Cystine stones

ALGORITHM

Dilute (urine ≥**3 L/day** including at night; sodium ≤2300 mg/day; moderate protein — not in children) → cystine <**250 mg/L** → **alkalinise** with potassium citrate to pH **7.0–7.5** → **thiol** (tiopronin)

- Cystine pKa 8.3 unreachable; pH >7.5 risks calcium phosphate. Sodium alkali raises cystine excretion.
- Thiols form a soluble cystine–drug disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

9 · Infection (struvite) stones

- **Surgical disease first**: complete clearance (percutaneous nephrolithotomy for staghorn → *Ch 98*); every fragment harbours bacteria.

- **Acetohydroxamic acid (AHA)**: urease inhibitor. AUA: only for residual/recurrent struvite after surgical options exhausted. Toxicity ~30%: haemolytic anaemia, **deep venous thrombosis** (most severe), tremor, headache. Contraindicated: creatinine >**2.5 mg/dL**, pregnancy.
- Most struvite formers also hypercalciuric/hypocitraturic → 24-hour urine.

10 · Monitoring

ALGORITHM

Start therapy → 24-hour urine within **6 months** → then at least yearly urine + blood for side effects → new stone: **re-analyse** → periodic imaging

11 · Landmark trials

Trial	Design	Result	Take-home
Borghi 1996	Randomised controlled trial (RCT), high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein/salt, normal calcium halved events vs low calcium	Never restrict calcium
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

100 Lower Urinary Tract Calculi

1 · Bladder calculi

- ~5% of urinary stones. Types: **migrant**, **primary** (endemic), **secondary**.

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; male ≫ female	Men >60 with outlet obstruction
Cause	Cereal diet low in animal protein and phosphate → hypophosphaturia	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless cause removed

- Foreign bodies (migrated intrauterine device, sutures, Foley balloon fragments, hair during clean intermittent catheterisation [CIC]) → **heterogeneous nucleation**. Drugs: triamterene, indinavir.

> **KEY POINT**
> Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

- Adults: **terminal haematuria** with lower urinary tract symptoms. Children: sandy urine, dysuria.

2 · Management of bladder stones

The route, not the stone, dictates modality.

Option	Indication / key point
Chemodissolution	Adjunct only; catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; poor in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice; risk urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction/closure); spares urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

- Cystolithotomy = intact removal; cystolitholapaxy = mechanical crushing.

3 · Special situations

- **BOO** in 45–79%. **Bladder stone secondary to BOO = absolute indication for benign prostatic hyperplasia (BPH) surgery** — but >30% infected and some have no BOO → **prove BOO**.
- **Diversion/augmentation**: hyperchloraemic acidosis → hypercalciuria; enteric hyperoxaluria; urease producers → struvite; **mucus** biofilm. Conduit with stomal stenosis → clear + revise stoma; **continent pouch → percutaneous** (protects continence); neobladder → transurethral, or percutaneous/open if large.
- **Spinal cord injury**: risk peaks **3 months** after injury; 15–30% within 10 years. **CIC beats indwelling catheters**; suprapubic = urethral. ≤4 cm transurethral laser; larger

percutaneous/open; ESWL for anaesthetic risk or autonomic dysreflexia.

- **Transplant**: **remove all exposed suture**. **Children**: small urethra → percutaneous cystolithotripsy for large stones, even infants.

4 · Urethral calculi

- 0.3–1% of stones; males; mostly **migratory** (calcium oxalate, acute). Primary minority: struvite, from stricture, diverticulum, hair-bearing graft. Commonest site: posterior urethra (up to 88%).
- **Acute painful retention**; posterior → perineal/rectal pain; anterior → penile-tip pain, palpable mass. ~98% radio-opaque.

Site	Treatment
Posterior	**Push back** to bladder → laser; else in situ
Anterior	**No pushback**; in situ holmium laser; urethrotomy for large
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction + diverticulectomy or stricture repair, same or staged

5 · Unusual sites

- **Preputial**: phimosis, poor hygiene; struvite → circumcision, **excised skin to histopathology**.
- **Prostatic**: calcium phosphate, in acini (endogenous) or from reflux (exogenous). Linked to BPH, **not cancer**. Transrectal ultrasound (TRUS) main tool.
- **Seminal vesicle**: haematospermia, painful ejaculation; **TRUS most accurate**; vesiculectomy or transurethral resection of ejaculatory ducts; seminal vesiculoscopy has fewer complications.

54 Management of Paediatric Kidney Stone Disease

1 · Epidemiology

- Early onset = < **24 years**; rising fastest in **adolescent girls**; ~50% recur within 3–5 years.

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%** — mostly idiopathic, on Randall's plaques	75–80%
Calcium phosphate (CaP) / uric acid	CaP commoner, uric acid rarer — more alkaline urine	≈ 5% each
Lifetime stakes	Chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); cumulative radiation	Shorter

- **High risk**: augmentation (≈ 1%/year), malabsorption, tube feeds, immobility.

KEY POINT
Stone formers get more CKD, hypertension, fracture; childhood is the bone-accrual window — seek low BMD early.

2 · Diagnosis and imaging

ALGORITHM
Suspected stone → **ultrasound (US)** grey-scale + colour Doppler → negative + high suspicion or hydronephrosis → **low-dose non-contrast computed tomography (NCCT)** → negative + low suspicion: stop.

- US: **twinkle artefact**; sensitivity > 70%, specificity > 95%; overestimates size 1–2 mm. NCCT ≈ 100% accurate; 40 mA for ≤ 50 kg.

3 · Metabolic evaluation — every child

Abnormality raises recurrence up to **fivefold** → **full evaluation after the first stone**.

- **History**: ketogenic diet, vitamins C/D, furosemide, acetazolamide, topiramate, prematurity.
- **Examination clues**: dysmorphism → Williams; rickets → Dent, hereditary hypophosphataemic rickets with hypercalciuria (HHRH); tetany → familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC).
- **Stone analysis** first: cystine (hexagonal crystals), struvite, 2,8-dihydroxyadenine, xanthine diagnostic.
- Parathyroid hormone only with hypercalcaemia + hypophosphataemia. Hypercalcaemia + hypercalciuria → vitamin D metabolite ratio for **CYP24A1** loss (mimics vitamin D intoxication). Proteinuric boys → urine β2-microglobulin (Dent).
- Not toilet-trained → spot solute : creatinine ratios.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m²/day
Citrate (lower limit)	Hypocitraturia < 128 mg/g creatinine (male), < 300 (female)

Metabolite	Normal (child)
Cystine	< 50–60 mg/1.73 m²/day
Uric acid	< 815 mg/1.73 m²/day; spot < 0.56 mg/dL glomerular filtration rate

KEY POINT
Never use adult limits: infants excrete more calcium, so the school-age 0.21 ratio overcalls them. Citrate's abnormality is too *little*.

4 · Urinary abnormalities

Hypercalciuria — commonest (30–50%)

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (calcium-sensing receptor gain)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH → else **idiopathic**

Hyperoxaluria

- P2 Primary Hyperoxaluria — *urinary glycolate/L-glycerate → genetics; pyridoxine*
- **Secondary**: fat malabsorption, low-calcium diet, vitamin C.

Others

- **Hypocitraturia**: incomplete dRTA needs acid-load test.
- **Cystinuria**: *SLC3A1*/*SLC7A9*; may present in infancy with staghorn.
- **Hyperuricosuria**: hypoxanthine–guanine phosphoribosyltransferase (HPRT) deficiency (Lesch–Nyhan), phosphoribosyl pyrophosphate synthetase (PRPS) superactivity, *SLC22A12*.

Monogenic disease	Inheritance	Signature
Dent (*CLCN5*)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (*CLDN16/19*)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · Management principles

- Off-label tamsulosin ≤ 6 weeks. Up to 60% need surgery.
- Residual fragment threshold: **4 mm on US** (2 mm CT + ≈ 2 mm US overestimate).
- Prophylaxis: all ureteroscopy (URS) and percutaneous nephrolithotomy (PCNL); shockwave lithotripsy (SWL) only if infection risk; ≤ 24 h.

Scenario (American Urological Association / Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation ± α-blocker
Ureteric, failed/unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting

Scenario (American Urological Association / Endourological Society 2016)	Recommendation
Renal ≤ **20 mm**	SWL or URS
Renal > 20 mm	PCNL or SWL (with stent/nephrostomy); authors: PCNL
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · Ureteroscopy

- Clearance ≈ 90%; worse < 6 years, < 20 kg, lower pole. French (Fr) sizes: semirigid 4.5–6, flexible 7.5–9.

ALGORITHM
Sterile culture → lithotomy → cystoscopy → retrograde pyelogram + safety wire → calibrate orifice with 8/10 Fr coaxial dilator → **if it will not pass: 4.6–4.8 Fr stent ≥ 14 days, return later** → URS → stent if long case, many passes, trauma/oedema.

- **No balloon or active orifice dilatation**; passive stent dilatation.

7 · Shockwave lithotripsy

- Short skin-to-stone distance; equals URS for small stones.
- Favourable: < 10 mm, non-lower-pole, < 1000 Hounsfield units. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- < 20 kV; ≤ 2000–3000 shocks (< 2000 infants); **80/min better than 120/min**.
- Absolute contraindications: bleeding diathesis, untreated UTI, pregnancy.

8 · Percutaneous nephrolithotomy

- First-line > 20 mm. 24–30 Fr in a child ≈ 72 Fr adult → **miniaturise** (mini < 20–24, micro < 10 Fr).

ALGORITHM
Prevent hypothermia → ureteric catheter → prone (latex precautions in myelomeningocele) → US/fluoroscopic puncture of posterior calyx → serial dilators → 18 Fr sheath (24 Fr staghorn) → nephroscope + ultrasonic/laser → flexible check → nephrostomy (7 Fr infant to 16 Fr adolescent).

- Transfusion < 10%; late bleeding → angioembolisation.

9 · Secondary prevention

- **Fluid**: ≥ maintenance; > 2–2.5 L/day adolescents; 1.5–2 L/m² cystinuria/PH.
- **Sodium** < 2–3 mEq/kg/day. **Calcium not restricted** (binds gut oxalate).
- Urine pH: > 6.5 uric acid, > 7 cystine; avoid > 6.5 in calcium formers.

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; hypokalaemia → add potassium citrate

Drug	Indication	Child dose / caution
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP if pH > 6.5
Tiopronin	Cystinuria failing fluid + alkali	Fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

Thiols and lumasiran are the only drugs approved for paediatric prevention. † PH1 with end-stage renal disease: combined liver–kidney transplantation; kidney-alone grafts fail.

10 · Landmark trials

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate ↓ 65%	RNA interference therapy for PH1

125 Genital and Lower Urinary Tract Trauma

1 · Principles

- Lower tract injuries are rarely life-threatening; long-term disability follows mismanagement, not the injury.
- Key decision — **immediate reconstruction vs deliberate delay**: testicular rupture → immediate repair; pelvic fracture urethral injury (PFUI) → suprapubic tube now, urethroplasty later.
- Imaging: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · Penile fracture

- P2 Fracture penis — *tunica thinnest ventrolaterally; snap → detumescence → eggplant deformity; deviates away from tear; urethral injury ~20% West vs ~3% Middle East; clinical diagnosis, ultrasound then magnetic resonance imaging (MRI); intraoperative flexible cystoscopy; delay ≤7 days harmless; complications ~20%*

> **ALGORITHM**
> Prompt exploration → ventral penoscrotal (or small lateral) incision → evacuate haematoma → identify tear (artificial erection) → irrigate → close tunica with interrupted **2-0/3-0 absorbable** → repair urethra if injured

- Bilateral tears (~10%) carry more urethral injury; absence of blood at meatus does not exclude it. Collagenase ruptures mostly after the 14-day observation window → conservative unless urethral injury.
- Breached Buck's fascia → †butterfly haematoma. A negative MRI reliably excludes fracture. Distal circumcising incision → add circumcision if uncircumcised.

3 · Penetrating, bite, amputation, zipper and strangulation

- **Gunshot**: explore immediately (do not delay for imaging) → irrigate → remove foreign matter → antibiotics → close tunica; artificial erection confirms closure. Almost never isolated. Urethral injury **15–50%** → RUG or intraoperative dye. Viable tissue → primary repair; tissue loss (high-velocity) → suprapubic diversion, staged repair.
- **Dog bites**: irrigate, debride, **primary closure**, amoxicillin–clavulanate; tetanus and rabies cover. **Human bites**: late, infected → **leave open**; amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- Usually self-mutilation → **psychiatric consultation always**. Preferred: **microsurgical replantation**; else macroscopic urethra/corpora repair.
- **"Double bag"**: saline gauze → sterile bag → outer bag with ice slush. Long ischaemia alone is no reason to abandon replantation.
- Erections >50% either technique; microvascular → less stricture, skin and sensory loss. Part unavailable → formalise stump; total loss → radial forearm phalloplasty.

> **ALGORITHM**
> Suprapubic cystostomy → debride → tunica 2-0 absorbable → urethra over catheter 4-0 → dorsal artery **11-0**, dorsal vein **9-0**, dorsal nerve **10-0** nylon → skin cover

- **Zipper** (children): penile block, mineral oil, one attempt; else cut cloth between teeth or the slider's **median bar** with bone cutters; circumcision if needed.
- **Strangulation**: children → hair tourniquet, consider **child abuse**. Lubricate and remove → **string technique** (tape under ring, wound distally, unwound from proximal end) → glanular puncture → cutting tools with saline cooling in theatre. Suprapubic catheter if retention.

4 · Testicular injury

- Protected by scrotal mobility, cremasteric reflex, tough tunica. ~80% of penetrating scrotal wounds involve neighbours (thigh, penis, urethra, **femoral vessels**).
- Consider rupture in every blunt injury; haematoma size does not predict it. **Ultrasonography** sensitivity only ~64–75% — **equivocal scan must not delay exploration**.
- Non-palpable testis after trauma → **dislocation**, commonest to superficial inguinal pouch (50%). Trauma does not exclude torsion.

†American Association for the Surgery of Trauma (AAST) testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM
Explore → remove necrotic/extruded tubules → close tunica with small absorbable sutures → large defect: tunica vaginalis flap/graft (no synthetic) → drain

- Explore **significant haematocele** (up to 80% hide rupture) and **intratesticular haematoma** (delayed orchidectomy 15%). Penetrating → always explore; ligate injured vas (non-absorbable), reconstruct later; consider contralateral exploration. Close even small defects (rising pressure extrudes tubules).
- Salvage **>90%** if repaired within **72 h**; non-operative care → infection, atrophy, late orchidectomy.

5 · Genital skin loss

- Causes: Fournier debridement (commonest), avulsion, burns, blast. Fournier: crepitus, subcutaneous air → antibiotics, resuscitation, **repeated debridement**, negative pressure wound therapy with daily inspection; early suprapubic diversion if extensive; hyperbaric oxygen unproven. Deep electrical burns: conservative.

	Penis	Scrotum
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch	**Meshed** STSG after local flaps
Key step	Excise residual subcoronal skin (lymphoedema); never reuse avulsed skin	Suture testes together (avoid bifid scrotum); ≤50% loss closes primarily
Note	Graft insensate, sexual function preserved	Thigh pouches not recommended

- Graft 5–7 days after last debridement (or ≥72 h culture-directed antibiotics, granulation, normal white count); bolster or vacuum dressing.

6 · Bladder injury

- P2 Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries — *pelvic fracture in 83–95% of bladder injuries, bladder in 5–10% of fractures; cystography for gross haematuria + pelvic fracture; retrograde fill **350 mL**, drainage film (plain only); flame-shaped extraperitoneal; 22-Fr catheter for uncomplicated extraperitoneal; intraperitoneal → immediate repair; suprapubic tube adds nothing*

†AAST bladder grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Iatrogenic: obstetric/gynaecological surgery commonest. Penetrating pelvic/buttock injury with any haematuria → cystography.
- Poor drainage = catheter problem until proven otherwise. Bone spicules in the wall prevent healing → repair.
- Intraperitoneal tears are larger than imaging suggests; leak → chemical peritonitis, **acidosis and azotaemia**. Laparoscopic repair in selected isolated cases. Orifice injury → stented reimplant.

> **KEY POINT**
> Immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, laparotomy for other reasons.

7 · Posterior urethral injury (PFUI)

- P2 P3 Lower urinary tract trauma in pelvic fracture / early and late management of PFUD — *bulbomembranous junction; triad blood at meatus, retention, palpable bladder; RUG technique; suprapubic tube standard, brief realignment optional (no difference); delayed excision and primary anastomosis at **2–3 months** (corporal split → pubectomy → rerouting); erectile dysfunction (ED) up to 82%, restenosis 5–15%, incontinence <4%*
- ~10% of male, up to 6% of female pelvic fractures; highest risk with **straddle (four-rami)** and unstable fractures. Sphincter usually preserved → continence often maintained. "High-riding" prostate unreliable.
- Females: vaginal examination in every pelvic fracture; urethroscopy preferred. Gap = **stenosis**, not stricture.
- Suprapubic tube safe with pelvic hardware (tunnel high in midline). Partial tear → one gentle catheter attempt. Realignment: catheter out 4–6 weeks, keep suprapubic tube. Endoscopic "cut-to-the-light" fails for gaps ≥1 cm. Limit lithotomy ≤5 h.

†AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation

†AAST urethra grade	Injury
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

ALGORITHM

Blood at meatus → RUG before catheter: 16-Fr catheter in fossa navicularis, balloon 3 mL → penile traction → oblique → 20 mL contrast → image during injection

- ED predictors: symphyseal diastasis, lateral displacement, long gap; consider penile duplex Doppler (ischaemia may predispose to restenosis).

8 · Anterior urethral injury

- Mostly **straddle injury** (bulbar urethra against pubis). Contusion/incomplete → catheter; major → suprapubic tube. Low-velocity gunshot → primary repair; high-velocity → diversion. Debride spongiosum minimally.
- Delayed: **excision and primary anastomosis** of the 1.5–2 cm bulbar segment (>95% success); urethrotomy through obliteration is futile.

9 · Landmark trials

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

62 Paediatric Genitourinary Trauma

1 · Renal injury: how the child differs

- Kidney = commonest site of paediatric genitourinary (GU) trauma; mostly **blunt**, grade 1–3; half have other injuries. Nephrectomy (3–5%) **more likely in adult than paediatric trauma units**. Congenital anomalies (e.g. ureteropelvic junction [UPJ] obstruction) 3–5 × commoner than in adults.

Feature	Child	Adult
Protection	Less perirenal fat, weaker wall, kidney below ribs	Better cushioned
Blood pressure	Maintained until ≈ **25–30% volume loss** — hypotension late	Falls earlier
Haematuria	May be absent (up to 2/3 of grade ≥ 2)	Classic trigger
Imaging decision	**Mechanism-led**; radiation → ALARA (as low as reasonably achievable)	Haematuria/hypotension-led

2 · Indications for imaging

Setting	Indication (American Urological Association [AUA])
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria + systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest — even without haematuria

- Paediatric triggers: fall > **10 feet (3 m)**, high-speed collision, flank blow; rib, spine, femur or pelvic fracture. All four markers together → > **98% sensitivity**.
- † Paediatric hypotension (1–10 years): SBP < 70 + (2 × age) mm Hg.

> **KEY POINT**
> Trap: "no haematuria, so no imaging". Haematuria is one of four routes into imaging; in children both adult markers fail, so mechanism drives the decision.

3 · Imaging studies

- FAST (focused sonography): free fluid only, not staging. Renal ultrasound (US): no radiation, detects grade 3–5; EAU (European Association of Urology) Paediatric: screening.
- **Contrast computed tomography (CT)** preferred. Children often get one phase → add a **delayed (excretory) phase** if a leak is suspected — it finds urine, not blood.

4 · Grading

American Association for the Surgery of Trauma (AAST): 1–3 = **low**; 4–5 = **high**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia

Grade	1989 AAST	2011 revision
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

KEY POINT

Grades 1–3 differ only in depth (1 cm) and never involve the collecting system — entry makes grade 4. Shattered kidney: grade 5 (1989), grade 4 (2011).

High-risk CT criteria (grade 4–5)

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation + no contrast in distal ureter	Urine leak needing intervention

0–1 factor → < 10% intervention; 2–3 → > 50% (grade 4a vs 4b).

5 · Non-operative management (NOM)

- **Default at every grade**, incl. grade 4–5 and stable penetrating injury, if stable (after resuscitation).

ALGORITHM

Confirm stability → grade → bed rest → serial haemoglobin + abdominal examination → mobilise when gross haematuria resolves and counts stable.

- Grade 1–2 → **23-hour observation**. Repeat imaging: AUA CT at 48 h (grade 4–5); EAU US at 2–3 days — but **symptoms, not imaging, predict complications**.
- Grade 3 (devitalised) to 5: ≈ **25%** embolisation, ≈ **15%** urinoma intervention, ≈ 5% exploration.

6 · Complications during NOM

- **Urine leak**: grade 4–5 only; up to **85%** resolve. Urinoma triad: **flank pain, ileus, low-grade fever**. No distal ureteric contrast → cystoscopy + retrograde pyelography (RPG). Escalate: **stent + Foley** (AUA first line, ≈ 90%; out 4–6 weeks) → percutaneous drain → nephrostomy → open repair.

Bleeding	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Fistula or pseudoaneurysm ruptures as haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild

Bleeding	Persistent	Delayed
Unstable	Urgent angioembolisation	

- Superselective embolisation > 80% success; retained function grade 4 ≈ 35–45%, grade 5 ≈ 10–30%.

7 · Operative management

- Indications: **instability from renal bleeding**; pulsatile/expanding haematoma; failed embolisation. Stable non-pulsatile haematoma at laparotomy → **leave alone**.
- **One-shot pyelogram**: 2 mL/kg → film at 10–15 min; confirms a **contralateral kidney** only.
- P2 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — *vascular control, renorrhaphy, nephrectomy; revascularise only solitary/bilateral; Page kidney*

> **ALGORITHM**
> Contralateral kidney confirmed → open Gerota → control vessels → debride → close collecting system → parenchyma over bolster → drain. Penetrating: inspect ureter.

- **UPJ disruption**: deceleration + hyperextension; †**Chance (lap-belt) fracture**, aortic, duodenal injury. Haematuria may be absent. CT: **medial extravasation, empty distal ureter, often no laceration** → RPG. Stable, early → stented ureteropelvic anastomosis; unstable or > 1 week → nephrostomy, repair at 2–3 months.
- Pre-existing UPJ obstruction → earlier stent, pyeloplasty 8–12 weeks. Duplex: upper-pole stent will not drain a lower-pole leak.

8 · Follow-up

- Scarring: grade 1–2 none; 3 ≈ 50–60%; 4–5 ≈ 100% → **US at 2–4 months**; dimercaptosuccinic acid (DMSA) scan if function loss suspected.
- **Hypertension** (> 30 days): rare, renin-mediated → renin–angiotensin blockade. Yearly blood pressure after grade 3–5. Solitary kidney: sport with padding; avoid motorised sport.

9 · Ureteral trauma

- Mostly penetrating. **Haematuria absent in 25–50%**; ≈ 10% missed at laparotomy. Diagnose by **delayed-phase CT** or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

- ≤ 1 week: stable → repair; unstable → ligate + nephrostomy. > 5–7 days: stent first; fails → nephrostomy, reconstruct at 6–12 weeks.
- P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — *repair by level: ureteroureterostomy, Boari, reimplant ± psoas hitch; ileal ureter, autotransplant*
- **Avoid transureteroureterostomy in children** (risks the other kidney).

10 · Bladder injury

Feature	Child	Adult
Bladder position	**More intraperitoneal** → isolated injury commoner (≈ half without fracture)	Pelvic organ
Pelvic fracture	Half as frequent → higher energy → more urethral injury	Commoner
Posterior urethral injury	Prostatic; **bladder neck injury commoner**	Bulbomembranous

- **Gross haematuria or retention** → **retrograde cystography** + urethrography (RUG); delayed-phase CT may miss rupture.
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V bladder neck or trigone.
- Extraperitoneal → catheter (repair if bone fragments, hardware, rectal/vaginal tear, bladder neck). Intraperitoneal → **always repair**. Catheter 1–2 weeks → cystogram.

KEY POINT
Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor — many need secondary procedures, even closure and diversion.

11 · Urethral injury

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- **RUG before any catheter**. Catheter passes → partial; stricture recurs in half after urethrotomy. Not passable → **suprapubic tube**. Realignment brief only; no immediate urethroplasty (EAU). **Bladder neck injury repaired at once.**

ALGORITHM
Suprapubic diversion → wait several months → cystogram + RUG ± endoscopy → transperineal excision and primary anastomosis (pubectomy if tension) → catheter 2–3 weeks → cystogram → suprapubic tube out.

- Restricture ≈ 10%, incontinence ≤ 20%, erectile dysfunction up to half (gap > 2.5 cm). Girls: ≈ 5% of pelvic fractures, with vaginal/rectal injury.

12 · Genital injury

KEY POINT
Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Examine anus and rectum; involve social services before discharge.

- Penis AAST: I skin; II Buck fascia; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or ≥ 2 cm; V total penectomy.
- **Circumcision** (commonest): worst with the **Mogen clamp** (glans amputation); lyse adhesions carefully *→ Ch 55*. **Hair tourniquet**: painless → fistula or amputation.
- Testis AAST: I contusion; II subclinical tunica tear; III < 50% loss; IV ≥ 50% loss; V destruction. US: tunica discontinuity, haematocele, focal flow loss → **explore and close tunica**; US at 6 weeks (masked tumour).
- Vulva: straddle ≈ 80%, labial. Hymen/vaginal/perianal injury = **penetration → abuse evaluation**; **normal examination does not exclude abuse**.

★ Putting it together

A · Renal colic

ALGORITHM

1. NSAID (ketorolac 30 mg) unless renal failure, bleeding, pregnancy. Non-contrast computed tomography (NCCT) confirms; ultrasound first in pregnancy and children (→ *Ch 96*).

2. Decompress now (stent or nephrostomy, equal) if infected, bilateral, solitary kidney, immunosuppressed, renal deterioration, refractory pain/vomiting. Never lithotripsy when infected.

3. Otherwise <10 mm → passage trial ± α-blocker (distal 5–10 mm) 4–6 weeks → fails: ureteroscopy (URS) or shock wave lithotripsy (SWL) (→ *Ch 97*).

B · Stone operation

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS clears more; SWL fewer complications. Hard proximal >15–20 mm → percutaneous nephrolithotomy (PCNL)
Renal <2 cm, not lower pole	SWL or URS	SWL best <900 Hounsfield units, skin-to-stone <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line (Lower Pole I)
Burden >2 cm, staghorn	PCNL	Staged URS an option

KEY POINT

SWL absolute contraindications: pregnancy, coagulopathy/antithrombotics, untreated infection, uncontrolled hypertension, arterial aneurysm, distal obstruction. No routine stent after SWL/uncomplicated URS (→ *Ch 98*).

C · Prevention

ALGORITHM

All: urine ≥2.5 L/day †, normal calcium, low sodium, less animal protein (→ *Ch 99*).

High risk → 24-hour urines → targeted therapy → repeat within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide + potassium citrate (hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate
Uric acid	Alkalinise to pH 6.5–7 †; allopurinol only for hyperuricosuria/gout or failed alkali
Cystine	≥3 L/day, pH 7.0–7.5, then tiopronin
Struvite	Complete clearance; check metabolism

D · Renal trauma

ALGORITHM

Image (delayed-phase CT): gross haematuria, microhaematuria + shock, deceleration, penetrating (→ *Ch 94*).

Stable grade I–IV, selected V → non-operative (>90%). Blush → embolise. Symptomatic leak → stent.

Explore if unstable or expanding haematoma; one-shot pyelogram (2 mL/kg, 10 min) confirms other kidney.

E · Bladder and urethra

Injury	Management
Extraperitoneal bladder rupture	22 Fr catheter, cystogram 2–3 weeks; repair if laparotomy, spicule, unhealed at 4 weeks
Intraperitoneal bladder rupture	† Surgical repair
Pelvic fracture urethral injury	Suprapubic tube (or brief realignment) → delayed urethroplasty
Penile fracture	Prompt tunical repair; check urethra (→ *Ch 125*)
Testicular rupture	Immediate repair

Cystography: retrograde, ≥350 mL.

F · Ureteric defects

Defect	Repair
2–3 cm, upper/mid	Spatulated ureteroureterostomy
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplant, nephrectomy (→ *Ch 93*)

G · Children

KEY POINT

Stones: ultrasound first; full metabolic evaluation after first stone. Renal ≤20 mm → SWL or URS; >20 mm → PCNL (→ *Ch 54*). **Trauma:** image by mechanism (fall >3 m, deceleration) — blood pressure falls late; perirenal haematoma >2.2 cm predicts embolisation (→ *Ch 62*).